Combinatorial Geometry

Labix

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Contents

1 Polyhedrons and Polytopes

1.1 Convex Geometry

Definition 1.1.1 (Convex Sets)
Let $C \subseteq \mathbb{R}^n$. We say that C is convex if for all $x, y \in C$ and $t \in [0, 1]$, we have $(1-t)x + ty \in C$.

Definition 1.1.2 (Convex Hull)
Let $S \subseteq \mathbb{R}^n$ be a subset. Define the convex hull of S to be

$$\text{Conv}(S) = \left\{ \sum_{k=1}^{n} \lambda_k x_k \;\middle|\; x_k \in S \text{ and } \sum_{k=1}^{n} \lambda_k = 1 \right\}$$

Proposition 1.1.3
Let $S \subseteq \mathbb{R}^n$ be a subset. Then we have

$$\text{Conv}(S) = \bigcap_{\substack{S \subseteq C \subseteq \mathbb{R}^n \\ C \text{ is convex}}} C$$

Theorem 1.1.4 (Caratheodory's Theorem)
Let $S \subseteq \mathbb{R}^n$ be a subset. Then there exists $x_1, \dots, x_{n+1} \in S$ such that

$$\text{Conv}(S) = \text{Conv}(x_1, \dots, x_{n+1})$$

1.2 Polyhedrons and its Properties

Polytopes exists more generically in Euclidean space. We lose nothing discussing polytopes in $\mathbb{R}^n$ since Euclidean spaces maybe identified with $\mathbb{R}^n$ canonically.

Definition 1.2.1 (Halfspaces)
A halfspace of $\mathbb{R}^n$ is a subset of $\mathbb{R}^n$ of the form

$$\{x \in \mathbb{R}^n \mid \langle a, x \rangle + b \geq 0\}$$

for some $a, b \in \mathbb{R}^n$.

Definition 1.2.2 (Polyhedrons)
Let $P \subseteq \mathbb{R}^n$. We say that P is a polyhedon if there exists finitely many half spaces $H_1, \dots, H_k$ such that

$$P = \bigcap_{i=1}^{k} H_i$$

Definition 1.2.3 (Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. We say that P is a polytope if P is bounded.

Theorem 1.2.4 (Weyl-Minkowski)
Let $P \subseteq \mathbb{R}^n$ be a subset. Then P is a polytope if and only if there exists a finite subset $S \subseteq \mathbb{R}^n$ such that $P = \text{Conv}(S)$.

Definition 1.2.5 (Dimension)
Let $P \subseteq \mathbb{R}^n$ be a polytope. Define the dimension of P to be the dimension of the smallest affine subspace containing P.

1.3 The Faces of a Polyhedron

Definition 1.3.1 (Supporting Hyperplane)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $H \subseteq \mathbb{R}^n$ be a hyperplane. We say that H is a supporting hyperplane for P if the following are true.
- $P \cap H \neq \emptyset$.
- P lies entirely in one of the halfspaces defined by H.

Definition 1.3.2 (Faces)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. A face of P is a subset of P defined by

$$F = P \cap H$$

where H is a supporting hyperplane for P.

Lemma 1.3.3
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $F \subseteq P$ be a face. Then F is a polytope.

Lemma 1.3.4
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $F, G \subseteq P$ be faces of P. Then $F \cap G$ is a face of P.

Lemma 1.3.5
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $F \subseteq P$ be a face of P. Let $G \subseteq F$ be a face of F. Then G is a face of P.

Definition 1.3.6 (Vertices, Edges and Facets)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $F \subseteq P$ be a face.
- We say that F is a vertex if $\dim(F) = 0$. Denote the set of vertices of P by $V(P)$.
- We say that F is an edge if $\dim(F) = 1$.
- We say that F is a facet if $\dim(F) = \dim(P) - 1$.

Definition 1.3.7 (The Face Poset)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Define the face poset $\mathcal{F}(P)$ of P to be the set of faces of P together with inclusion.

Definition 1.3.8 (The Face Vector)
Let $P \subseteq \mathbb{R}^n$ be a polytope of dimension d. Define the face vector $(f_0, \dots, f_{d-1})$ of S by the formula

$$f_i = \text{The number of } i\text{-dimensional faces of } P$$

for $0 \leq i \leq d-1$ and by convention, $f_{-1} = f_d = 1$.

Lemma 1.3.9 (Euler's Formula)
Let $P \subseteq \mathbb{R}^n$ be a polytope of dimension d. Then we have

$$\sum_{i=0}^{d-1} (-1)^i f_i = 1 + (-1)^d$$

Definition 1.3.10 (Combinatorially Equivalent)
Let $P_1 \subseteq \mathbb{R}^{n_1}$ and $P_2 \subseteq \mathbb{R}^{n_2}$ be polytopes. We say that P_1 and P_2 are combinatorially equivalent if there exists a inclusion preserving bijection between $\mathcal{F}(P_1)$ and $\mathcal{F}(P_2)$. In this case we write $P_1 \simeq P_2$.

Definition 1.3.11 (Edge Graph)
Let $P \subseteq \mathbb{R}^n$ be a polytope. Define the edge graph $G(P)$ of P by the following data.
- $V(G(P)) = V(P)$.
- $E(G(P)) = \{(p, q) \mid \text{ There exists an edge } E \text{ such that } \partial E = \{p, q\}\}$.

1.4 Dual Polytopes

Definition 1.4.1 (Polar Dual)
Let $P \subseteq \mathbb{R}^n$ be a polytope. Define the polar dual of P to be

$$P^* = \{y \in \mathbb{R}^n \mid \langle x, y \rangle \leq 1 \text{ for all } x \in P\}$$

Proposition 1.4.2
Let $P \subseteq \mathbb{R}^n$ be a polytope. Suppose that $0 \in P$. Then the following are true.
- P^* is a polytope.
- $P = (P^*)^*$.
- Let $F \subseteq P$ be a face. Then F^* is a face of P^* of dimension $\dim(P) - 1 - \dim(F)$.
- There is a inclusion reversing bijection $\mathcal{F}(P) \overset{1:1}{\leftrightarrow} \mathcal{F}(P)^{\text{op}}$ given by $F \mapsto F^*$.

Definition 1.4.3 (Dual Graph)
Let $P \subseteq \mathbb{R}^n$ be a polytope. Define the dual graph $G^*(P)$ as follows.
- $V(G^*(P)) = \{F \subseteq P \mid F \text{ is a facet of } P\}$.
- $E(G^*(P)) = \{(F, G) \mid F \cap G \text{ is a face of dimension } \dim(P) - 2\}$.

Lemma 1.4.4
Let $P \subseteq \mathbb{R}^n$ be a polytope such that $0 \in P$. Then $G(P^*) = G^*(P)$.

2 Operations on Polytopes

2.1 Minkowski Sum

Definition 2.1.1 (Minkowski Sum)
Let $P, Q \subseteq \mathbb{R}^n$ be polytopes. Define the Minkowski sum of P and Q by

$$P + Q = \{x + y \in \mathbb{R}^n \mid x \in P, y \in Q\}$$

Proposition 2.1.2
Let $P, Q \subseteq \mathbb{R}^n$ be polytopes. Suppose that $P = \text{Conv}(v_1, \dots, v_p)$ and $Q = \text{Conv}(w_1, \dots, w_q)$. Then $P + Q$ is a polytope and

$$P + Q = \text{Conv}\left(\{v_i + w_j \mid 1 \leq i \leq p, 1 \leq j \leq q\}\right)$$

2.2 The Join of Polytopes

Definition 2.2.1 (The Join of Two Polytopes)
Let $P \subseteq \mathbb{R}^n$ and $Q \subseteq \mathbb{R}^m$ be polytopes. Define the join of P and Q as follows. Embed P into $\mathbb{R}^{n+m+1}$ by $x \mapsto (x, 0, \dots, 0)$ and embed Q into $\mathbb{R}^{n+m+1}$ by $y \mapsto (0, \dots, 0, y, 1)$. Then the join of P and Q is

$$P * Q = \{(1 - \lambda)x + \lambda y \mid x \in P, y \in Q, 0 \leq \lambda \leq 1\}$$

Proposition 2.2.2
Let P and Q be polytopes. Then the following are true.

- $\dim(P * Q) = \dim(P) + \dim(Q) + 1$.
- $\mathcal{F}(P * Q) = \mathcal{F}(P) * \mathcal{F}(Q)$.
- $V(P * Q) = V(P) \amalg V(Q)$.

Example 2.2.3
Let $n, m \in \mathbb{N}^+$. Then we have

$$\Delta^n * \Delta^m = \Delta^{n+m+1}$$

Example 2.2.4
Let P be a polytope. Then the following are true.

- $P * \Delta^1 \cong \Sigma P$.
- $P * \Delta^0 \cong SP$.

3 Symmetries of a Polytope

3.1 Congruent Polytopes

Definition 3.1.1 (Congruent Polytopes)
Let $P \subseteq \mathbb{R}^n$ and $Q \subseteq \mathbb{R}^m$ be polytopes. We say that P and Q are congruent if there exists a function $\varphi : P \to Q$ such that the following are true.

- φ is a bijection.
- $\|\varphi(y) - \varphi(x)\| = \|y - x\|$ for all $x, y \in P$.

Proposition 3.1.2
Let $P \subseteq \mathbb{R}^n$ and $Q \subseteq \mathbb{R}^m$ be polytopes. Then P and Q are congruent if and only if there exists an isometry $\varphi : \mathbb{R}^n \to \mathbb{R}^m$ such that $\varphi(P) = Q$.

Definition 3.1.3 (Symmetries of a Polytope)
Let $P \subseteq \mathbb{R}^n$ be a polytope. A symmetry of P is a congruence $\varphi : P \to P$. Denote the group of symmetries of P by

$$\text{Sym}(P) = \{\varphi : P \to P \mid \varphi \text{ is a congruence }\} \leq \text{Isom}(\mathbb{R}^n)$$

Lemma 3.1.4
Let $P \subseteq \mathbb{R}^n$ be a polytope. Let $F \subseteq P$ be a face of P. Let $\varphi : P \to P$ be a symmetry. Then $\varphi(F)$ is a face of dimension $\dim(F)$.

Definition 3.1.5 (Vertex Centroid of a Polytope)

Proposition 3.1.6
Let $P \subseteq \mathbb{R}^n$ be a polytope. Let $\tau : \mathbb{R}^n \to \mathbb{R}^n$ by the map defined by $\tau(x) = x - \overline{P}$. Then there is an isomorphism

$$\text{Sym}(P) \cong \tau\text{Sym}(P)\tau^{-1}$$

WLOG we may translate any polytope P such that $\overline{P} = 0 \in \mathbb{R}^n$.

Proposition 3.1.7
Let $P \subseteq \mathbb{R}^n$ be a polytope such that $\overline{P} = 0$. Then we have

$$\text{Sym}(P) \leq O(\text{AffSpan}(P)) \cong O(\dim(P), \mathbb{R})$$

3.2 Affine Isomorphisms

Definition 3.2.1 (Affinely Isomorphic)
Let $P_1 \subseteq \mathbb{R}^{n_1}$ and $P_2 \subseteq \mathbb{R}^{n_2}$ be polytopes. We say that P_1 and P_2 are affinely isomorphic if there exists an affine transformation $f : \mathbb{R}^{n_1} \to \mathbb{R}^{n_2}$ such that $f|_{P_1}$ is a bijection.

Note that f itself need not be an affine isomorphism (i.e. an isomorphism of affine spaces).

Definition 3.2.2 (Affine Symmetries of a Polytope)
Let $P \subseteq \mathbb{R}^n$ be a polytope. An affine symmetry of P is an affine isomorphism $f : P \to P$. Denote the group of affine symmetries of P by

$$\text{Aff}(P) = \{f : P \to P \mid f \text{ is an affine isomorphism }\} \leq \text{Aff}(\mathbb{R}^n)$$

Proposition 3.2.3
Let $P\subseteq\mathbb{R}^n$ be a polytope. Let $\tau:\mathbb{R}^n\to\mathbb{R}^n$ by the map defined by $\tau(x)=x-\overline{P}$. Then there is an isomorphism
$$\text{Aff}(P)\cong\tau\text{Aff}(P)\tau^{-1}$$

Proposition 3.2.4
Let $P\subseteq\mathbb{R}^n$ be a polytope such that $\overline{P}=0$. Then we have
$$\text{Aff}(P)\leq O(\text{AffSpan}(P))\cong\text{GL}(\dim(P),\mathbb{R})$$

Proposition 3.2.5
Let $P\subseteq\mathbb{R}^n$ be a polytope. Then we have
$$\text{Sym}(P)\leq\text{Aff}(P)\leq\text{Aut}(\mathcal{F}(P))$$

4 The Duality between Simple and Simplicial Polytopes

4.1 Simplicial Polytopes

Definition 4.1.1 (Simplicial Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope. We say that P is a simplicial polytope if all its facets are simplicies.

Lemma 4.1.2
Let $P \subseteq \mathbb{R}^n$ be a simplicial polytope. Then ∂P is a simplicial complex.

Lemma 4.1.3
Let $P \subseteq \mathbb{R}^n$ be a simplicial polytope. Then ∂P is a triangulation of $S^{\dim(P)-1}$.

Proposition 4.1.4 (The Dehn-Sommerville Equation)
Let $P \subseteq \mathbb{R}^n$ be a simplicial polytope of dimension d. Then the following are true:

$$\sum_{j=k}^{d-1}(-1)^j\binom{j+1}{k+1}f_j = (-1)^{d-1}f_k$$

for $-1 \leq k \leq d-2$.

In particular, ∂P is a simplicial complex. The notion of face vector coincides, and they enjoy both the Dehn-Sommerville equation and its relation with the h-vector.

Example 4.1.5
Substituting $k = -1$ in the Dehn-Somemerville equation recovers Euler's identity.

Example 4.1.6
Let P be a simplicial 3-polytope. Then the Dehn-Sommmerville equations give

$$2 = f_0 - f_1 + f_2 \quad \text{and} \quad 2f_1 = 3f_2$$

Substituting gives $f_1 = 3f_0 - 6$. Hence the f-vector of P is given by

$$f(P) = (f_0, 3f_0 - 6, 2f_0 - 4)$$

Example 4.1.7
Let P be a simplicial 4-polytope. Then the Dehn-Sommmerville equations give

$$0 = f_0 - f_1 + f_2 - f_3 \quad \text{and} \quad 2f_0 + 3f_2 = 2f_1 + 4f_3 \quad \text{and} \quad 2f_3 = f_2$$

Substituting gives $f_1 = 3f_0 - 6$. Hence the f-vector of P is given by

$$f(P) = (f_0, f_1, 2(f_1 - f_0), f_1 - f_0)$$

4.2 Simple Polytopes

Definition 4.2.1 (Simple Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope. We say that P is a simple polytope if for all vertices $v \in P$, there exists n edges $e_1, \dots, e_n \subseteq P$ such that $v \in \partial e_i$ for all i.

Lemma 4.2.2
Let $P_1 \subseteq \mathbb{R}^n$ and $P_2 \subseteq \mathbb{R}^m$ be simple polytopes. Then the Cartesian product $P_1 \times P_2$ is a simple polytope.

4.3 The Duality Theorem

Proposition 4.3.1
Let $P \subseteq \mathbb{R}^n$ be a polytope. Then P is simple if and only P^* is simplicial.

Duality of d-dimensional faces with $(n-d)$-dimensional faces.

Proposition 4.3.2
Let $P \subseteq \mathbb{R}^n$ be a polytope. If P is both simple and simplicial, then P belongs to one of the following types of polytopes.

- $n = 2$ (and so P is a polygon).
- P is a simplex.

4.4 The Nerve Complex

Definition 4.4.1 (The Nerve Complex)
Let $P \subseteq \mathbb{R}^n$ be a simple polytope. Define the nerve complex of P to be the simplicial complex

$$\mathcal{K}_P = \partial(P^*)$$

Lemma 4.4.2
Let $P_1 \subseteq \mathbb{R}^n$ and $P_2 \subseteq \mathbb{R}^m$ be simple polytopes. Then we have

$$\mathcal{K}_{P_1 \times P_2} \cong \mathcal{K}_{P_1} * \mathcal{K}_{P_2}$$

5 Extremal Polytopes

5.1 Cyclic Polytopes

Definition 5.1.1 (Cyclic Polytopes)
Let $\gamma : \mathbb{R} \to \mathbb{R}^d$ be the function $\gamma(t) = (t, \dots, t^d)$. Let $t_1, \dots, t_n \in \mathbb{R}$ such that $t_i < t_{i+1}$ for $1 \leq i \leq n-1$. Define the cyclic polytope of dimension d with n vertices by

$$C(n, d) = \text{Conv}(\{\gamma(t_1), \dots, \gamma(t_n)\})$$

Note: different choices of $t_1, \dots, t_n$ lead to combinatorially equivalent cyclic polytopes.

Lemma 5.1.2
Let $n, d \in \mathbb{N}^+$. Then $C(n, d)$ is a simplicial polytope.

Theorem 5.1.3 (Gale Evenness Condition)
Let $n, d \in \mathbb{N}^+$. Let $t_1, \dots, t_n \in \mathbb{R}$ such that

$$C(n, d) = \text{Conv}(\{\gamma(t_1), \dots, \gamma(t_n)\})$$

Then $\text{Conv}(\{t_{i_1}, \dots, t_{i_d}\})$ is a facet of $C(n, d)$ if and only if $i_k - i_j$ is an odd number for $1 \leq j, k \leq d$.

Proposition 5.1.4
Let $n, d \in \mathbb{N}^+$. Then we have

$$f_{\lfloor d/2 \rfloor - 1}(C(n, d)) = \binom{n - \lceil d/2 \rceil}{\lfloor d/2 \rfloor} + \binom{n - \lfloor d/2 \rfloor - 1}{\lceil d/2 \rceil - 1}$$

Lemma 5.1.5
Let $n, d \in \mathbb{N}^+$. Then we have

$$f_i(C(n, d)) = \binom{n}{i+1}$$

for $0 \leq i \leq \left\lfloor \frac{n}{2} \right\rfloor - 1$.

Corollary 5.1.6
Let $n, d \in \mathbb{N}^+$. Then we have

$$f_i(C(n, d)) = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \binom{k}{d-1-i} \binom{n-d+k-1}{k} + \sum_{j=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{d-j}{d-1-i} \binom{n-d+j-1}{j}$$

for $0 \leq i \leq d$, where we set $\binom{p}{q} = 0$ if $p < q$ or $q < 0$.

Theorem 5.1.7 (Upper Bound Theorem)
Let P be a polytope of dimension d. Suppose that $f_0(P) = n$. Then we have

$$f_k(P) \leq f_k(C(n, d))$$

for all k.

5.2 Stacked Polytopes

Definition 5.2.1 (Vertex Pulling)
Let $P \subseteq \mathbb{R}^n$ be a polytope. Let $F \subseteq P$ be a face of P. Let $v \in \mathbb{R}^n \setminus P$ such that the following are true.

- Suppose that H is a supporting hyperplane of F. Then v and P are on different sides in $\mathbb{R}^n \setminus H$.
- Suppose that $G \subseteq P$ is a facet such that $G \neq F$, and H is a supporting hyperplane of G. Then v and P are on the same side $\mathbb{R}^n \setminus H$.

Define the pulling of P by v along F to be the polytope

$$P' = \text{Conv}(P \cup \{v\})$$

Definition 5.2.2 (Stacked Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope. We say that P is stacked if there exists a sequence of polytopes $\Delta^{\dim(P)} = S_0, \dots, S_k = P$ such that S_{i+1} is obtained from S_i by vertex pulling for all $0 \leq i \leq k-1$.

Proposition 5.2.3
Let $P \subseteq \mathbb{R}^n$ be a stacked polytope. Then the following are true.

- P is obtained from $\Delta^{\dim(P)}$ by $|V(P)| - \dim(P) - 1$ vertex pullings.
- P is simplicial.
- The f-vector of P is given by

$$f_k(P) = \binom{\dim(P)}{k} V(P) - \binom{\dim(P)+1}{k+1} k$$

  for $1 \leq k \leq \dim(P) - 1$.

Proposition 5.2.4
Let $P \subseteq \mathbb{R}^n$ be a simplicial polytope such that $\dim(P) \geq 2$. Then the following are equivalent.

- P is a stacked polytope.
- $G^*(P)$ is a tree.
- If $F \in \text{MF}(\partial P)$ is a missing face, then $\dim(F) = \left\lfloor \frac{\dim(P)}{2} \right\rfloor$.

When $\dim(P) = 3$, then P is a stacked polytope if and only if ∂P is a flag complex.

Theorem 5.2.5 (Lower Bound Theorem)
Let P be a simplicial polytope such that $\dim(P) \geq 3$. Suppose that $f_0(P) = n$. Let Q be a stacked polytope with n vertices. Then we have

$$f_k(P) \geq f_k(Q)$$

for all k.

5.3 Neighbourly Polytopes

6 Classical Families of Polytopes

6.1 Cross Polytopes

6.2 Regular Polytopes

Definition 6.2.1 (Flags of a Polytope)
Let $P \subseteq \mathbb{R}^n$ be a polytope. A flag of P is a maximal chain in $\mathcal{F}(P)$. Denote the set of flags of P by

$$\text{Flag}(P) = \{\Phi \subseteq \mathcal{F}(P) \mid \Phi \text{ is a flag of } P\}$$

Since symmetries of a polytope sends faces to faces of the same dimension, we obtain a group action of $\text{Sym}(P)$ on $\text{Flag}(P)$.

Definition 6.2.2 (Regular Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope. We say P is a regular polytope if $\text{Sym}(P)$ acts transitively on $\text{Flag}(P)$.

Lemma 6.2.3 | Let $P \subseteq \mathbb{R}^n$ be a regular polytope. Then P^* is regular.

Example 6.2.4 (Regular n-Polygons)
Let $n \geq 3$. Define the regular n-gon to be

$$P_n = \text{Conv}\left(\left\{\left(\cos\left(\frac{2\pi k}{n}\right), \sin\left(\frac{2\pi k}{n}\right)\right) \;\middle|\; 0 \leq k \leq n-1\right\}\right)$$

Then there is an isomorphism

$$\text{Sym}(P_n) \cong D_{2n}$$

Proposition 6.2.5
Let P be a regular polytope of dimension 1. Then P is congruent to $[-1, 1]$.

Proposition 6.2.6
Let P be a regular polytope of dimension 2. Then P is congruent to P_n for some $n \geq 3$.

Proposition 6.2.7
Let P be a regular polytope of dimension 3. Then P is congruent to exactly one of the following.
- The tetrahedron.
- The cube.
- The octahedron.
- The dodecahedron.
- The icosahedron.

7 Piecewise Linear Structure

7.1 Relation between Polyhedra and Simplicial Complexes

Definition 7.1.1 (Locally Finite Collections)
Let X be a space. Let $P \subseteq \mathcal{P}(X)$ be a collection of subsets. We say that P is locally finite if for each $p \in X$, there exists a neighbourhood $U \subseteq X$ of p such that the set

$$\{V \in P \mid U \cap V \neq \emptyset\}$$

is finite.

Proposition 7.1.2
Let P be a polyhedra. Then P is a locally finite union of simplexes.

Thus if P is a polytope, then P has the underlying structure of a simplicial complex.

Proposition 7.1.3
Let $P \subseteq \mathbb{R}^n$ be a polytope. Then there is a PL homeomorphism

$$P \cong_{\text{PL}} D^{\dim(P)}$$

7.2 Piecewise Linear Maps between Polyhedra

Proposition 7.2.1
Let P, Q be polyhedra. Let $f : P \to Q$ be a function. Then the following are equivalent.

- f is piecewise linear.
- For each $p \in P$, there exists $L \subseteq P$ such that $f(\lambda p + \mu x) = \lambda f(p) + \mu f(x)$ for all $\lambda, \mu \geq 0$ and $\lambda + \mu = 1$ and $x \in L$.
- $\Gamma(f) = \{(x, f(x)) \mid x \in P\}$ is a polyhedron.
- There exists a locally finite decomposition into polyhedra $P = \bigcup_{\alpha \in I} P_\alpha$ such that for each $\alpha \in I$, the restriction $f|_{P_\alpha}$ is affine linear.

Lemma 7.2.2
Let P, Q be polyhedra. Let $f : P \to Q$ be a piecewise linear function. If f is a homeomorphism, then f is a piecewise linear homeomorphism.

7.3 Polytopal Spheres

Definition 7.3.1 (Polytopal Spheres)
Let S be a simplicial complex. We say that S is a polytopal sphere if S is the boundary of a polytope.

Together with results from Simplicial Topology, we have a series of inclusions

$$\text{Polytopal Spheres} \subseteq \text{Combinatorial Spheres} \subseteq \text{Simplicial Spheres}$$

When $\dim = 2$, all inclusions are equalities. When $\dim = 3$, the first inclusion is strict and the second inclusion is an equality. At $\dim = 4$, the second inclusion is an open problem, at $\dim \geq 5$, the inclusion is strict. If the number of vertices is $\leq \dim + 4$, then all inclusions are equalities.

8 Cones and Fans

8.1 Cones and its Properties

Definition 8.1.1 (Cones)
Let $C \subseteq \mathbb{R}^n$. We say that C is a cone if $x \in C$ implies that $sx \in C$ for all $s \geq 0$.

Definition 8.1.2 (Polyhedral Cones)
A cone in $\mathbb{R}^n$ is a subset $C \subseteq \mathbb{R}^n$ such that if $c_1, \dots, c_n \in C$, then $\sum_{i=1}^n \lambda_i c_i \in C$ for all $\lambda_1, \dots, \lambda_n \geq 0$.

Definition 8.1.3 (Conical Hull)
Let $S \subseteq \mathbb{R}^n$ be finite. Define the conical hull of S to be

$$\text{Cone}(S) = \left\{ \sum_{s \in S} \lambda_s s \;\middle|\; \lambda_s \geq 0 \right\}$$

Proposition 8.1.4
Let $C \subseteq \mathbb{R}^n$ be a subset. Then the following are equivalent.
- C is a polyhedral cone.
- There exists a finite subset $S \subseteq \mathbb{R}^n$ such that $C = \text{Cone}(S)$.
- C is the intersection of finitely many halfspaces containing 0 in their boundary.

Therefore cones are also polyhedrons.

Definition 8.1.5 (Dimension of a Cone)
Let $C \subseteq \mathbb{R}^n$ be a cone. Define the dimension of C to be the dimension of the smallest affine subspace containing C.

Definition 8.1.6 (Rational Cones)
Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is rational if there exists a finite set $S \subseteq \mathbb{Z}^n$ such that $C = \text{Cone}(S)$.

Definition 8.1.7 (Regular Cones)
Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is regular if there exists a finite set $S \subseteq \mathbb{Z}^n$ such that the following are true.
- $C = \text{Cone}(S)$.
- Elements of S are $\mathbb{Z}$-linearly independent.

Definition 8.1.8 (Simplicial Cones)
Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is simplicial if there exists a finite set $S \subseteq \mathbb{R}^n$ such that the following are true.
- $C = \text{Cone}(S)$.
- Elements of S are $\mathbb{R}$-linearly independent.

8.2 The Dual Cone

Definition 8.2.1 (The Dual of a Cone)
Let $C \in \mathbb{R}^n$ be a cone. Define the dual of C to be

$$C^\vee = \{x \in \mathbb{R}^n \mid \langle x, c \rangle \geq 0 \text{ for all } c \in C\}$$

Definition 8.2.2 (Associated Hyperplane and Halfspace of a Linear Functional)
Let $\phi \in (\mathbb{R}^n)^*$. Define the associated hyperplane of ϕ to be

$$H_\phi = \{x \in \mathbb{R}^n \mid \phi(x) = 0\}$$

Define the associated halfspace of ϕ to be

$$H_\phi^+ = \{x \in \mathbb{R}^n \mid \phi(x) \geq 0\}$$

Lemma 8.2.3
Let $C \in \mathbb{R}^n$ be a cone. Let H be a hyperplane in $\mathbb{R}^n$. Then H is a supporting hyperplane of C if and only if there exists $\phi \in C^\vee$ such that $H = H_\phi$.

Lemma 8.2.4
Let $C \in \mathbb{R}^n$ be a polyhedral cone. Then the following are true.
- $C^\vee$ is a polyhedral cone.
- $(C^\vee)^\vee = C$.
- $C = \bigcap_{i=1}^n H_{\phi_i}$ for $\phi_i \in (\mathbb{R}^n)^*$ if and only if $C^\vee = \text{Cone}(\phi_1, \dots, \phi_n)$.
- If C is strongly convex, then $C^\vee$ is strongly convex.

Lemma 8.2.5 (Separation Lemma)
Let $C_1, C_2 \subseteq \mathbb{R}^n$ be cones such that $F = C_1 \cap C_2$ is a face of both C_1 and C_2. Then there exists $\phi \in C_1^\vee \cap (-C_2)^\vee$ such that

$$F = C_1 \cap H_\phi = C_2 \cap H_\phi$$

Definition 8.2.6 (Strongly Convex Cones)
Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is strongly convex if C does not contain a line.

Lemma 8.2.7
Let $C \subseteq \mathbb{R}^n$ be a cone. Then the following are equivalent.
- C is strongly convex.
- $\{0\}$ is a face of C.
- $C \cap (-C) = \{0\}$.
- $\dim(C^\vee) = n$.

8.3 Fans and its Properties

Definition 8.3.1 (Fans)
Let $n \in \mathbb{N}^+$. A fan in $\mathbb{R}^n$ is a finite set of cones Σ such that the following are true.
- Each $C \in \Sigma$ is a strongly convex rational polyhedral cone.
- For each cone $C \in \Sigma$ and each face F of C, we have $F \in \Sigma$.
- For cones $C_1, C_2 \in \Sigma$, we have $C_1 \cap C_2$ are faces of C_1 and C_2.

We write $\Sigma(d) = \{C \in \Sigma \mid \dim(C) = d\}$ for the d-dimensional cones in the fan.

Definition 8.3.2 (The Support of a Fan)
Let $n \in \mathbb{N}^+$. Let Σ be a fan in $\mathbb{R}^n$. Define the support of Σ to be

$$|\Sigma| = \cup_{C \in \Sigma} C$$

Definition 8.3.3 (Types of Fans)

Definition 8.3.4 (Complete Fans)
Let $n \in \mathbb{N}^+$. Let Σ be a fan in $\mathbb{R}^n$. We say that Σ is complete if $|\Sigma| = \mathbb{R}^n$.

8.4 Normal Fan